

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10: 1.3 THE PERIODIC TABLE

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have finished all 10 lessons on the Periodic Table. Now write your own complete explanation of the driving question: How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form? Use your Element Cards, your notes from each lesson, the Driving Question Board (DQB) and your Class Model (the Electron Arrangement Model) from Lesson 1 to Lesson 10. Write in complete sentences, use scientific words correctly (atomic number, outer electrons, group, period, noble gas, cation, anion, valency, oxidation number, radical), and give evidence from the activities to support every claim. Show your reasoning clearly: say what you saw, then explain why. Write your answers in the space under each prompt. Take your time and show what you have learned.

Part 1: How Is the Periodic Table Ordered?

In Lesson 1 you sorted the first 20 Element Cards by mass and by properties. (a) Explain how Mendeleev (1869) and Moseley (1913) each ordered the elements. (b) Use the masses and atomic numbers of Ar and K from your Element Cards, and what you know about how each reacts, to explain why the modern table is ordered the way it is. (c) Define a group and a period, and describe where Li, Na and K and where Ne and Ar fell when you arranged the cards by atomic number. (d) Look at the masses of Li, Na and K on your cards, and at Ca and Ar. Why was mass alone not enough to explain the patterns you saw?

(a) Mendeleev in 1869 arranged the elements by increasing atomic mass and left gaps for elements that had not yet been discovered. Moseley in 1913 showed that atomic number is the true basis of order. (b) Ar has Z 18 and A_r 40, while K has Z 19 and A_r 39. If we ordered by mass, K would come before Ar. But Ar behaves like Ne (unreactive) and K behaves like Li and Na (reactive), so they only fit their families when ordered by atomic number. This is why the modern table uses atomic number. (c) A group is a vertical column and a period is a horizontal row. When I arranged the cards by atomic number into 4 periods, Li, Na and K fell in one column (group 1), and Ne and Ar fell together in the noble gas column (group 18). (d) Li, Na and K have very different masses (7, 23 and 39) but behave alike, and Ca has the same A_r as Ar (40) but sits in a different group. So mass cannot explain the pattern. Something hidden inside the atom must decide where an element belongs and how it behaves, and this led us to electron arrangement.

Part 2: Electron Arrangement, Group and Period

(a) Write the electron arrangement of each of these atoms in both energy-level form and s and p notation: Li, Na, Cl, Ca. (b) State the rule that links the electron arrangement to the group number for groups 1 and 2 and for groups 13 to 18, and the rule that links it to the period number. Say what is special about He. (c) Use these rules to state the group and period of Na, Cl and Ne. (d) Use your answers to explain why Li, Na and K sit in the same column although their masses are very different, and why neighbours such as Na and Mg do not behave alike.

(a) Li: 2,1 and $1s^2 2s^1$. Na: 2,8,1 and $1s^2 2s^2 2p^6 3s^1$. Cl: 2,8,7 and $1s^2 2s^2 2p^6 3s^2 3p^5$. Ca: 2,8,8,2 and $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$. (b) The number of outer electrons decides the group: for groups 1 and 2 the group number equals the number of outer electrons, and for groups 13 to 18 the group number is the number of outer electrons plus 10 (so groups 13 to 18 have 3 to 8 outer electrons). The number of occupied energy levels is the period number. He ($1s^2$) is a special case: it has 2 outer electrons, which fill its only energy level, so it is placed in group 18 with the noble gases rather than in group 2. (c) Na has 1 outer electron and 3 occupied energy levels, so it is in group 1, period 3. Cl has 7 outer electrons and 3 occupied levels, so it is in group 17 ($7 + 10$), period 3. Ne (2,8) has 8 outer electrons and 2 occupied levels, so it is in group 18 ($8 + 10$), period 2. (d) Li (2,1), Na (2,8,1) and K (2,8,8,1) each have 1 outer electron, so they are all in group 1 even though their masses are 7, 23 and 39. Na has 1 outer electron but Mg (2,8,2) has 2, so they are in different groups and behave differently. Electron arrangement, which follows from atomic number, is the hidden property that decides the position of an element in the periodic table.

Part 3: Families, Stability and Reactivity

(a) Name the family of each: Li, Mg, Cl and Ar, and give the number of outer electrons of each family. (b) Describe what you saw when Li, Na and K were added to water, and state the reactivity trend down group 1. (c) Use the electron arrangements of Ne (2,8), Ar (2,8,8), Na and Cl to explain why Ne and Ar do not react while Na reacts vigorously and chlorine is used for water treatment in Nairobi and Mombasa.

(a) Li is an alkali metal (group 1, 1 outer electron). Mg is an alkaline earth metal (group 2, 2 outer electrons). Cl is a halogen (group 17, 7 outer electrons). Ar is a noble gas (group 18, full outer level of 8 electrons). (b) Li fizzed steadily and floated. Na melted into a ball, darted about and fizzed fast. K melted, burned with a lilac flame and was very vigorous. Reactivity increases down the group: $Li < Na < K$. (c) Ne and Ar have full outer levels (an octet), which is a stable arrangement, so they do not react. Na has only 1 outer electron, so by losing it, it reaches the stable arrangement of Ne (2,8). It loses 1 electron instead of gaining 7 because that is far easier. Cl has 7 outer electrons, so gaining just 1 electron gives it the Ar arrangement (2,8,8), which is why chlorine reacts easily and can be used to disinfect water in Nairobi and Mombasa. Atoms react to reach a full outer level like a noble gas.

Part 4: Predicting Ions, Valency and Formulae

(a) For each of Na, Mg, Al, O and Cl, write the ion formed with its charge and its electron

(a) Na^+ is $1s^2 2s^2 2p^6$ and matches Ne. Mg^{2+} is $1s^2 2s^2 2p^6$ and matches Ne. Al^{3+} is $1s^2 2s^2 2p^6$ and matches Ne. O^{2-} is $1s^2 2s^2 2p^6$ and matches Ne. Cl^- is $1s^2 2s^2 2p^6 3s^2 3p^6$ and matches Ar. (b) Na +1, Mg +2, Al +3, Cl -1, O -2. Valency is the number of electrons an atom loses, gains or shares and is always a positive whole number, while the oxidation number is the same number

arrangement in s and p notation, and state which noble gas arrangement it matches. (b) What is the oxidation number of Na, Mg, Al, Cl and O? Explain how it differs from valency. (c) Use the criss-cross method to derive the formulae of calcium chloride, aluminium oxide, and the fertilizer made from potassium and the nitrate radical, used on farms in Kericho and the Rift Valley. Use your Radical Cards from Lesson 8. (d) Why can outer electron count alone not predict the ion of iron or copper? Use the compounds from Lesson 7 as evidence.	with a sign: + for loss and - for gain. (c) Calcium chloride: Ca^{2+} and Cl^-, criss-cross gives CaCl_2. Aluminium oxide: Al^{3+} and O^{2-} gives Al_2O_3. Potassium nitrate: K^+ and the nitrate radical NO_3^- give KNO_3. In each, the total positive charge equals the total negative charge so the compound is neutral. (d) Iron and copper are transition elements that can lose different numbers of electrons. Iron forms Fe^{2+} in FeCl_2 and Fe^{3+} in Fe_2O_3, so it has two oxidation numbers (+2 and +3). Copper forms Cu^{2+} in CuO and can also form Cu^+, so it also has more than one oxidation number (+1 and +2). The outer-electron rule alone therefore does not predict a single oxidation number for these elements.
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Part 5: Putting It All Together and Predicting Strontium

Strontium (Sr, Z38) is an element you have never worked with. Its electron arrangement is 2,8,18,8,2. (a) Use this arrangement and the rules you have built to predict its group and period, and explain how you decided. (b) Predict the ion it forms and the formulae of its compounds with chlorine and oxygen. (c) Write a balanced equation with state symbols for sodium reacting with water, and explain why potassium and lithium follow the same pattern. (d) Write a final explanation of the driving question: how does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds they form?	(a) Sr has 2 outer electrons (the last number in 2,8,18,8,2) and 5 occupied energy levels (five numbers), so it is in group 2, period 5. (b) It loses 2 electrons to form Sr^{2+}, with a noble gas arrangement like Mg^{2+} and Ca^{2+}. Criss-cross with Cl^- gives SrCl_2 and with O^{2-} gives SrO. (c) $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$. Li and K each have 1 outer electron like Na, so each loses it to form a 1+ ion and follows the same pattern: $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$ and $2\text{Li} + 2\text{H}_2\text{O} \rightarrow 2\text{LiOH} + \text{H}_2$. The pattern depends on the outer electron count, not on mass (7, 23 and 39). (d) Elements are ordered by atomic number, as Moseley showed, and atomic number gives the electron arrangement. The number of outer electrons decides the group and the number of occupied energy levels decides the period. Elements in the same group, such as Li, Na and K, have the same outer electron count so they react alike despite different masses, while neighbours such as Na, Mg, Ne and Ar differ because their outer electron counts differ. Atoms lose or gain electrons to reach the stable noble gas arrangement. This is why Ne and Ar are unreactive, group 1 metals form 1+ ions and chlorine forms Cl^-. Knowing the ion charges lets us predict valencies and formulae such as NaCl, CaCl_2, Al_2O_3 and KNO_3 for any element, including ones we have never worked with. How atoms share electrons is explained in Sub-Strand 1.4, Chemical Bonding.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Ordering and history of the periodic table	Correctly explains Mendeleev (mass, gaps) and Moseley (atomic number), and uses the Ar and K data to justify ordering by atomic number.	Describes both scientists correctly but gives limited use of the Ar and K evidence.	Mixes up the scientists or the basis of order, or gives no evidence.
Electron arrangement, group and period	Writes correct arrangements in both forms and applies the group and period rules accurately (including the He exception) to explain why Li, Na and K are alike and Na and Mg differ.	Writes mostly correct arrangements and states the rules, with small errors in applying them.	Arrangements or rules are incorrect or not linked to group and period.
Stability and reactivity reasoning	Uses full outer levels, losing and gaining electrons, and specific evidence (water reactions, Ne, Ar, Cl) in a clear claim-evidence-reasoning explanation.	Explains stability and reactivity with some evidence but reasoning is incomplete.	Describes observations without linking them to electron arrangement.
Predicting ions, valency and formulae	Correctly gives ions with their arrangements, oxidation numbers, formulae, a balanced equation with state symbols, and explains the limits for iron and copper and the prediction for strontium.	Mostly correct predictions with minor errors in charges, formulae or balancing.	Many incorrect ions, formulae or an unbalanced equation.
Scientific communication	Complete sentences, accurate vocabulary and Kenyan examples, and a coherent final explanation that answers the driving question.	Mostly clear with correct vocabulary but some gaps in the final explanation.	Unclear writing, vocabulary misused, final explanation missing key ideas.

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